

STABLE MATCHING AND A TWO-TO-ONE MAP FOR TUNNELL'S TERNARY FORMS

ABSTRACT. For every odd positive squarefree congruent number n , we construct a deterministic map $\Phi_n : \mathcal{B}_n \rightarrow \mathcal{A}_n$ whose fibers have cardinality two, where $\mathcal{B}_n$ and $\mathcal{A}_n$ are the representation sets of n by $2x^2 + y^2 + 8z^2$ and $2x^2 + y^2 + 32z^2$. Tunnell's theorem supplies the identity $|\mathcal{B}_n| = 2|\mathcal{A}_n|$; more generally, the construction applies to every odd positive squarefree n satisfying this identity. Halving the third coordinate gives the even branch. On the odd branch, three integral quarter-turns define a partial bijection. We complete it by passing the residual points to their antipodal orbits and applying stable matching, with preferences determined by primitive midpoint directions. Equal-rank edges are disjoint, so the stable matching is unique. Its preference lists can be generated arithmetically from rank-two lattice cosets, and the proposal counts form a parking function with total at most $m(m+1)/2$, where m is the number of residual orbits on either side.

1. INTRODUCTION

Tunnell in [9] related the congruent-number problem to representation numbers of ternary quadratic forms. In this paper, we use his counting identity to construct a two-to-one map between the corresponding representation sets. For an odd positive integer n , we set

$$\begin{aligned} (1) \quad \mathcal{B}_n &= \{(x, y, z) \in \mathbb{Z}^3 : 2x^2 + y^2 + 8z^2 = n\}, \\ (2) \quad \mathcal{A}_n &= \{(u, v, w) \in \mathbb{Z}^3 : 2u^2 + v^2 + 32w^2 = n\}. \end{aligned}$$

A ternary quadratic form is a homogeneous quadratic polynomial in three variables, and a representation of n is simply an integer triple at which the form has value n . We call n *squarefree* if no prime square divides it. Both sets above are finite because each squared coordinate is bounded by n .

A positive integer is *congruent* if it is the area of a right triangle with rational side lengths. Equivalently, the elliptic curve $y^2 = x^3 - n^2x$ has a rational point of infinite order; see [6, Chap. I, §9] and [9, pp. 323–324]. A consequence of Tunnell's theorem is that every odd positive squarefree congruent number satisfies

$$(3) \quad |\mathcal{B}_n| = 2|\mathcal{A}_n|.$$

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We use this identity from Tunnell's theorem. The construction below does not require any further modular-form or elliptic-curve arguments.

Example 1.1 (The balance for $n = 41$). For $n = 41$, a solution in $\mathcal{B}_{41}$ must satisfy $|x| \leq 4$, $|y| \leq 6$, and $|z| \leq 2$; a solution in $\mathcal{A}_{41}$ must satisfy the same first two bounds and $|w| \leq 1$. Sorting by the last coordinate gives

$\mathcal{B}_{41} : z$	0	1	-1	2	-2	total
number of (x, y)	4	8	8	6	6	32

and

$\mathcal{A}_{41} : w$	0	1	-1	total
number of (u, v)	4	6	6	16

Thus $|\mathcal{B}_{41}| = 2|\mathcal{A}_{41}|$. For example,

$$(2, -1, 2) \in \mathcal{B}_{41}, \quad (2, -1, 1) \in \mathcal{A}_{41},$$

and the even-branch map simply halves the last coordinate.

Bach and Ryan [2] give methods for computing the representation numbers in this identity. Here we take (3) as given and ask how to pair the individual representations. We will work with any odd positive squarefree n satisfying this identity, so the construction is not restricted to the congruent-number case. We do not reprove Tunnell's identity or give a new congruent-number criterion.

Separate $\mathcal{B}_n$ according to the parity of its third coordinate:

$$\begin{aligned} \mathcal{B}_n^{\text{even}} &= \{(x, y, z) \in \mathcal{B}_n : z \equiv 0 \pmod{2}\}, \\ \mathcal{B}_n^{\text{odd}} &= \{(x, y, z) \in \mathcal{B}_n : z \equiv 1 \pmod{2}\}. \end{aligned}$$

The map

$$(4) \quad (x, y, 2w) \longmapsto (x, y, w)$$

is a bijection $\mathcal{B}_n^{\text{even}} \xrightarrow{\sim} \mathcal{A}_n$. Consequently (3) is equivalent to $|\mathcal{B}_n^{\text{odd}}| = |\mathcal{A}_n|$, and the essential problem is to construct a bijection from the odd branch to the target set.

Three quarter-turns give a bijection on part of $\mathcal{B}_n^{\text{odd}}$. We call the points outside its domain and image the residual points. To match them, we pass to their $\{\pm 1\}$ -orbits and join every source orbit to every target orbit. A primitive midpoint direction assigns a rank to each edge. Both endpoints order their incident edges by this rank, while equal-rank edges are vertex-disjoint. This is an instance of stable marriage with globally ranked pairs in the sense of Abraham, Levavi, Manlove, and O'Malley [1]. We will show that the stable matching is unique.

Stable matching and parking functions are classical objects [4, 5, 7, 8]. In our setting, the midpoint directions determine the preferences, and the numbers of proposals form a parking function. The counting identity alone would allow an arbitrary pairing. Our map is instead determined by the quarter-turns, the midpoint directions, and the orientation conventions fixed

below. It commutes with $X \mapsto -X$, and the same matching determines its inverse.

Theorem 1.2 (A two-to-one map). *Let n be an odd positive squarefree integer satisfying $|\mathcal{B}_n| = 2|\mathcal{A}_n|$. Fix the orientation and ordering conventions introduced below. Then the construction defines a deterministic map*

$$\Phi_n : \mathcal{B}_n \longrightarrow \mathcal{A}_n$$

such that every element of $\mathcal{A}_n$ has exactly two preimages. On $\mathcal{B}_n^{\text{even}}$, the map is (4). On $\mathcal{B}_n^{\text{odd}}$, it is the disjoint union of three quarter-turn branches and the unique stable matching of the remaining orbit sets. The algorithms below compute this matching from arithmetically generated preference lists, without first storing its complete edge table.

Here a *fiber* is the set of all points mapping to one target. The map comes from two bijections. The first gives the even preimage of each target, and the second gives the odd preimage:

$$(5) \quad \mathcal{B}_n^{\text{even}} \xrightarrow[\text{bijection}]{(x,y,2w) \mapsto (x,y,w)} \mathcal{A}_n, \\ \mathcal{B}_n^{\text{odd}} = D_n \sqcup \mathcal{B}_{n,\circ}^{\text{odd}} \xrightarrow[\text{bijection}]{\Lambda_n \sqcup \Psi_n^\circ} I_n \sqcup \mathcal{A}_{n,\circ} = \mathcal{A}_n.$$

The residual bijection is obtained by changing coordinates, matching antipodal orbits, and lifting back to points:

$$(6) \quad \mathcal{B}_{n,\circ}^{\text{odd}} \xrightarrow{\iota_B} \mathcal{S}_n^\circ \longrightarrow \overline{\mathcal{S}}_n \xrightarrow{\overline{M}_n} \overline{\mathcal{T}}_n \longrightarrow \mathcal{T}_n^\circ \xrightarrow{\iota_A^{-1}} \mathcal{A}_{n,\circ}.$$

The unlabeled arrows pass to antipodal orbits and then lift back with a specified sign. The cardinality hypothesis is used to prove that $\overline{M}_n$ is perfect; it is not used to choose the edge ranks.

Theorem 1.3 (Congruent-number case). *For every odd positive squarefree congruent number n , the construction defines a deterministic map $\Phi_n : \mathcal{B}_n \rightarrow \mathcal{A}_n$ such that every element of $\mathcal{A}_n$ has exactly two preimages.*

Section 2 introduces the two coordinate models, and Section 3 constructs the direct branches. Section 4 constructs the residual matching, and Section 5 defines the two-to-one map. Sections 6–8 explain how to compute the matching and work out the residual example at $n = 41$. Section 9 bounds the number of proposals and the cost of deferred acceptance. We finish with two questions about the residual map.

We will use two examples at $n = 41$. The first uses a quarter-turn, and the second uses the residual matching:

case	odd source	target	even source
direct quarter-turn	$(-4, -1, 1)$	$(2, -1, 1)$	$(2, -1, 2)$
residual matching	$(-2, -5, 1)$	$(4, -3, 0)$	$(4, -3, 0)$

Each row lists the two preimages of one target. The quarter-turns and the residual matching apply to disjoint subsets of $\mathcal{B}_n^{\text{odd}}$.

2. TWO CHANGES OF COORDINATES

We now introduce two changes of coordinates. The first is used for the quarter-turns and the second for midpoint directions. Each is applied to the original representation sets:

$$\begin{array}{ccccc} \mathcal{B}_n^{\text{odd}} & \xleftarrow[e_B^{-1}]{e_B} & L_1(n) & \xrightarrow{J_1, J_2, J_3} & L_0(n) \xleftarrow[e_A]{e_A^{-1}} \mathcal{A}_n, \\ \mathcal{B}_n^{\text{odd}} & \xleftarrow[\iota_B^{-1}]{\iota_B} & \mathcal{S}_n & \xleftarrow{\text{midpoint lines}} & \mathcal{T}_n \xleftarrow[\iota_A]{\iota_A^{-1}} \mathcal{A}_n. \end{array}$$

In the first row, both sets are represented by sums of three squares, where we can apply coordinate rotations. In the second, both lie on the same quadratic surface, and their half-sums and half-differences are integral.

The following table records the notation used below:

D_n, I_n, Λ_n	direct odd-source domain, its target image, and the quarter-turn bijection between them;
$\mathcal{B}_{n,\circ}^{\text{odd}}, \mathcal{A}_{n,\circ}, \Psi_n^\circ$	points left after the quarter-turn bijection and their residual bijection;
$\mathcal{S}_n, \mathcal{T}_n$	midpoint-coordinate lifts of the odd source and target sets;
$\overline{\mathcal{S}}_n, \overline{\mathcal{T}}_n$	antipodal residual orbit sets;
$\mathcal{P}_Q$	canonically oriented primitive midpoint directions;
Ψ_n, Φ_n	the full odd-branch bijection and the final two-to-one map.

2.1. Sum-of-three-squares coordinates for the direct map. For $\epsilon \in \{0, 1\}$, put

$$L_\epsilon(n) = \{(a, b, c) \in \mathbb{Z}^3 : a^2 + b^2 + c^2 = n, \quad a \equiv b \pmod{4}, \quad \frac{a-b}{4} \equiv \epsilon \pmod{2}\}.$$

Lemma 2.1 (Sum-of-three-squares coordinates). *The maps*

$$e_B(x, y, z) = (x + 2z, x - 2z, y), \quad e_A(u, v, w) = (u + 4w, u - 4w, v)$$

are bijections

$$e_B : \mathcal{B}_n^{\text{odd}} \xrightarrow{\sim} L_1(n), \quad e_A : \mathcal{A}_n \xrightarrow{\sim} L_0(n),$$

with inverses

$$\begin{aligned} e_B^{-1}(a, b, c) &= \left(\frac{a+b}{2}, c, \frac{a-b}{4} \right), \\ e_A^{-1}(a, b, c) &= \left(\frac{a+b}{2}, c, \frac{a-b}{8} \right). \end{aligned}$$

Proof. The norm identities are immediate:

$$(x + 2z)^2 + (x - 2z)^2 + y^2 = 2x^2 + y^2 + 8z^2,$$

$$(u + 4w)^2 + (u - 4w)^2 + v^2 = 2u^2 + v^2 + 32w^2.$$

The congruence conditions record whether $(a-b)/4$ is odd or even. Conversely, if $(a, b, c) \in L_1(n)$, then $(a+b)/2$ and $(a-b)/4$ are integral, the latter is odd, and the first norm identity gives a point of $\mathcal{B}_n^{\text{odd}}$. If $(a, b, c) \in L_0(n)$, then $(a-b)/8$ is integral and the second identity gives a point of $\mathcal{A}_n$. The displayed inverse formulas therefore give inverses on both sides. $\square$

Example 2.2 (The two coordinate changes at $n = 41$). Consider the odd source $X = (-4, -1, 1) \in \mathcal{B}_{41}^{\text{odd}}$. Lemma 2.1 sends it to

$$e_B(X) = (-2, -6, -1), \quad (-2)^2 + (-6)^2 + (-1)^2 = 41.$$

Here $(-2) - (-6) = 4$, so $(a-b)/4 = 1$ is odd, as required for $L_1(41)$. Likewise, for $Y = (2, -1, 1) \in \mathcal{A}_{41}$,

$$e_A(Y) = (6, -2, -1), \quad 6^2 + (-2)^2 + (-1)^2 = 41,$$

and $(6 - (-2))/4 = 2$ is even, as required for $L_0(41)$. The inverse formulas recover X and Y from these triples.

2.2. Midpoint coordinates for the residual graph. Put

$$Q(X_1, X_2, X_3) = 2X_1^2 + X_2^2 + 2X_3^2$$

and write

$$\langle X, Y \rangle_Q = 2X_1Y_1 + X_2Y_2 + 2X_3Y_3$$

for its associated bilinear form. Thus $Q(X+Y) = Q(X) + 2\langle X, Y \rangle_Q + Q(Y)$. Define

$$\mathcal{S}_n = \{E \in \mathbb{Z}^3 : Q(E) = n, E_3 \equiv 2 \pmod{4}\},$$

$$\mathcal{T}_n = \{F \in \mathbb{Z}^3 : Q(F) = n, F_3 \equiv 0 \pmod{4}\}.$$

The lifts

$$\iota_B(x, y, z) = (x, y, 2z), \quad \iota_A(u, v, w) = (u, v, 4w)$$

identify $\mathcal{B}_n^{\text{odd}}$ with $\mathcal{S}_n$ and $\mathcal{A}_n$ with $\mathcal{T}_n$.

Lemma 2.3 (Midpoint parity). *If $E \in \mathcal{S}_n$ and $F \in \mathcal{T}_n$, then*

$$\frac{E+F}{2}, \quad \frac{E-F}{2}$$

are integral vectors with odd third coordinate.

Proof. Since n is odd, the second coordinates of both E and F are odd. Their third coordinates are congruent to 2 and 0 modulo 4, respectively. Consequently, modulo 8 one has

$$n \equiv 2E_1^2 + 1 \equiv 2F_1^2 + 1,$$

so E_1 and F_1 have the same parity. The second-coordinate halves are integral because both entries are odd, and the third-coordinate halves are odd because $E_3 \equiv 2 \pmod{4}$ and $F_3 \equiv 0 \pmod{4}$. Thus both displayed vectors are integral and have odd third coordinate. $\square$

For $0 \neq v \in \mathbb{Z}^3$, define

$$\text{cont}(v) = \gcd(|v_1|, |v_2|, |v_3|).$$

Call v *primitive* if $\text{cont}(v) = 1$. Let $\mathcal{P}_Q$ contain the unique representative, with first nonzero coordinate positive, of every pair $\{d, -d\}$ of primitive vectors with d_3 odd. Order $\mathcal{P}_Q$ lexicographically by

$$(7) \quad \text{key}(d) = (Q(d), d_1, d_2, d_3).$$

For directions we choose the sign with first nonzero coordinate positive. For antipodal point orbits, introduced below, we instead choose the lexicographically smaller point.

Lemma 2.4 (Primitive generator of the midpoint line). *For every $E \in \mathcal{S}_n$ and $F \in \mathcal{T}_n$, there are unique $d \in \mathcal{P}_Q$ and a nonzero odd integer q such that*

$$\frac{E + F}{2} = qd.$$

They satisfy

$$\langle E, d \rangle_Q = qQ(d), \quad F = -E + 2qd.$$

For a fixed d , the edges $E \leftrightarrow F$ obtained in this way are pairwise disjoint.

Proof. By Lemma 2.3, the midpoint is a nonzero integral vector with odd third coordinate. Dividing by its content and applying the orientation convention gives q and d , with q odd. If $h = (E - F)/2$, equality of the norms gives $\langle d, h \rangle_Q = 0$, and hence $\langle E, d \rangle_Q = qQ(d)$ and $F = -E + 2qd$. Uniqueness follows from the unique sign-normalized primitive generator of $\mathbb{Q}(E + F) \cap \mathbb{Z}^3$, the integer points on the rational line spanned by $E + F$. For fixed E and d , the identity $q = \langle E, d \rangle_Q / Q(d)$ determines q , and then the formula determines F . Interchanging the roles of the endpoints gives the same conclusion at a target. Hence the edges belonging to a fixed direction are pairwise disjoint. $\square$

Example 2.5 (A midpoint direction at $n = 41$). Take the lifted residual points

$$E = (-2, -5, 2) \in \mathcal{S}_{41}, \quad F = (4, -3, 0) \in \mathcal{T}_{41}.$$

Both have Q -value 41. Their half-sum and half-difference are

$$\frac{E + F}{2} = (1, -4, 1), \quad \frac{E - F}{2} = (-3, -1, 1).$$

Both vectors are integral and have odd third coordinate, as in Lemma 2.3. The first vector is already primitive and canonically oriented, hence $q = 1$ and $d = (1, -4, 1)$. Moreover,

$$Q(d) = 20, \quad \langle E, d \rangle_Q = 20 = qQ(d), \quad -E + 2d = F.$$

The other midpoint line is generated, after sign normalization, by $e = (3, 1, -1)$, whose key begins with $Q(e) = 21$. Thus the key of d is the smaller of the two. When we pass to antipodal orbits, this comparison will determine which sign of the target is used.

3. A DIRECT QUARTER-TURN CORRESPONDENCE

Define three integral rotations in the sum-of-three-squares model by

$$J_1(a, b, c) = (a, -c, b), \quad J_2(a, b, c) = (c, b, -a), \quad J_3(a, b, c) = (-b, a, c).$$

Each is an element of $\text{SO}_3(\mathbb{Z})$: it is an integer matrix that preserves squared length and orientation. Geometrically, J_i fixes one coordinate axis and rotates the other two coordinates through a right angle.

Proposition 3.1 (Quarter-turn domains). *Let $(a, b, c) \in L_1(n)$. Then*

$$\begin{aligned} J_1(a, b, c) \in L_0(n) &\iff a + c \equiv 0 \pmod{8}, \\ J_2(a, b, c) \in L_0(n) &\iff c - b \equiv 0 \pmod{8}, \\ J_3(a, b, c) \in L_0(n) &\iff a + b \equiv 0 \pmod{8}. \end{aligned}$$

At most one of these three congruences holds.

Proof. The differences between the first two coordinates of the three images are $a + c$, $c - b$, and $-(a + b)$, respectively. Membership in $L_0(n)$ is therefore equivalent to the stated divisibility by 8.

In original odd-source coordinates, $c = y$ is odd and $a - b = 4z \equiv 4 \pmod{8}$. If any two of the displayed congruences held, then, after eliminating a or b , one would obtain

$$a - b \equiv -2c \pmod{8}.$$

Since c is odd, the right-hand side is 2 or 6 modulo 8, contradicting $a - b \equiv 4 \pmod{8}$. Hence the three domains are pairwise disjoint. $\square$

Writing the rotations in the original coordinates gives the following partial map. For $(x, y, z) \in \mathcal{B}_n^{\text{odd}}$ with $x + 2z + y \equiv 0 \pmod{8}$, set

$$(8) \quad \Lambda_1(x, y, z) = \left(\frac{x + 2z - y}{2}, x - 2z, \frac{x + 2z + y}{8} \right).$$

If $y - x + 2z \equiv 0 \pmod{8}$, set

$$(9) \quad \Lambda_2(x, y, z) = \left(\frac{x - 2z + y}{2}, -x - 2z, \frac{y - x + 2z}{8} \right).$$

Finally, if $x \equiv 0 \pmod{4}$, set

$$(10) \quad \Lambda_3(x, y, z) = \left(2z, y, -\frac{x}{4} \right).$$

The domains are pairwise disjoint by Proposition 3.1.

Theorem 3.2 (Direct partial bijection). *The formulas (8)–(10) define a partial bijection*

$$\Lambda_n : D_n \xrightarrow{\sim} I_n$$

from a subset $D_n \subseteq \mathcal{B}_n^{\text{odd}}$ to a subset $I_n \subseteq \mathcal{A}_n$. Its inverse is determined from a target $(u, v, w) \in \mathcal{A}_n$ by the following mutually exclusive conditions and formulas:

$$\begin{aligned} u + 4w - v \equiv 4 \pmod{8} : & \left(\frac{u + 4w + v}{2}, -u + 4w, \frac{u + 4w - v}{4} \right), \\ u - 4w + v \equiv 4 \pmod{8} : & \left(\frac{u - 4w - v}{2}, u + 4w, \frac{4w - u - v}{4} \right), \\ u \equiv 2 \pmod{4} : & \left(-4w, v, \frac{u}{2} \right). \end{aligned}$$

Moreover, Λ_n is equivariant under the antipodal involution $X \mapsto -X$.

Proof. The construction is $e_A^{-1} J_i e_B$ on each branch, so norm preservation and integrality follow from Proposition 3.1. The inverse formulas are $e_B^{-1} J_i^{-1} e_A$. Their conditions state exactly that the inverse image lies in $L_1(n)$. They are mutually exclusive: the first two cannot both hold because v is odd, and either of them is incompatible with $u \equiv 2 \pmod{4}$ by parity. Direct composition gives the two identity maps. Negation equivariance follows because each J_i is linear and each domain condition is unchanged by negation. $\square$

Example 3.3 (The three quarter-turn branches). The first two branches already occur at $n = 11$; the direct construction does not require the cardinality hypothesis. Direct substitution gives

branch	X	$e_B(X)$	$\Lambda_i(X)$
Λ_1	$(-1, -1, 1)$	$(1, -3, -1)$	$(1, -3, 0)$
Λ_2	$(-1, 1, -1)$	$(-3, 1, 1)$	$(1, 3, 0)$.

For the first row, $a + c = 1 - 1 = 0$, so J_1 applies and $J_1(1, -3, -1) = (1, 1, -3) = e_A(1, -3, 0)$. For the second row, $c - b = 1 - 1 = 0$, so J_2 applies and $J_2(-3, 1, 1) = (1, 1, 3) = e_A(1, 3, 0)$.

For $n = 41$, return to $X = (-4, -1, 1)$ from Example 2.2. Its sum-of-three-squares coordinates are $(a, b, c) = (-2, -6, -1)$. Since

$$a + b = -8 \equiv 0 \pmod{8},$$

Proposition 3.1 selects J_3 and neither of the other two quarter-turns. Indeed,

$$J_3(-2, -6, -1) = (6, -2, -1) = e_A(2, -1, 1).$$

In the original coordinates this is exactly

$$\Lambda_3(-4, -1, 1) = (2, -1, 1).$$

The inverse condition is $u \equiv 2 \pmod{4}$, and the third inverse formula gives $(-4w, v, u/2) = (-4, -1, 1)$.

Set

$$\mathcal{B}_{n,\circ}^{\text{odd}} = \mathcal{B}_n^{\text{odd}} \setminus D_n, \quad \mathcal{A}_{n,\circ} = \mathcal{A}_n \setminus I_n,$$

and let $\mathcal{S}_n^\circ = \iota_B(\mathcal{B}_{n,\circ}^{\text{odd}})$ and $\mathcal{T}_n^\circ = \iota_A(\mathcal{A}_{n,\circ})$. Under the hypothesis of Theorem 1.2,

$$(11) \quad |\mathcal{S}_n^\circ| = |\mathcal{T}_n^\circ|.$$

Indeed, $|\mathcal{B}_n^{\text{odd}}| = |\mathcal{A}_n|$ and Λ_n removes equally many points from the two sides.

4. STABLE MATCHING ON ANTIPODAL ORBITS

The residual sets are invariant under the antipodal involution $X \mapsto -X$. Since $n > 0$, no point in either set equals its negative, so each orbit has two elements. Define the orbit sets

$$\overline{\mathcal{S}}_n = \mathcal{S}_n^\circ / \{\pm 1\}, \quad \overline{\mathcal{T}}_n = \mathcal{T}_n^\circ / \{\pm 1\}.$$

By (11), these sets have the same cardinality; write

$$|\overline{\mathcal{S}}_n| = |\overline{\mathcal{T}}_n| = m.$$

Each orbit is an antipodal pair $[X] = \{X, -X\}$. Choose the lexicographically smaller of X and $-X$ as its canonical representative $\text{pc}([X])$. Let

$$G_n = K_{\overline{\mathcal{S}}_n, \overline{\mathcal{T}}_n}$$

be the complete bipartite graph on the two orbit sets.

Fix $[E] \in \overline{\mathcal{S}}_n$ and $[F] \in \overline{\mathcal{T}}_n$, with canonical representatives E and F . Apply Lemma 2.4 to the two edges between chosen representatives $E \leftrightarrow F$ and $E \leftrightarrow -F$. Let their direction keys be $k_+(E, F)$ and $k_-(E, F)$.

Lemma 4.1 (Edge ranking on antipodal orbits). *The two keys $k_+(E, F)$ and $k_-(E, F)$ are distinct. The quantity*

$$\kappa([E], [F]) = \min\{k_+(E, F), k_-(E, F)\}$$

is independent of the chosen representatives. There is a unique sign

$$\eta([E], [F]) \in \{\pm 1\}$$

such that the edge between representatives

$$E \longleftrightarrow \eta([E], [F])F$$

has key $\kappa([E], [F])$.

Proof. Write

$$\frac{E + F}{2} = qd, \quad \frac{E - F}{2} = ce$$

with primitive oriented directions d, e . Equality of the norms gives $\langle d, e \rangle_Q = 0$. The vectors are nonzero, so positive definiteness implies $d \neq e$ and hence the keys are distinct. Replacing either representative by its negative interchanges the two signed midpoint lines, so their unordered pair of keys is unchanged. The smaller one determines the unique sign. $\square$

Example 4.2 (Choosing the sign of a target). Take the canonical representatives

$$E = (-2, -5, -2), \quad F = (-4, -3, 0)$$

of the orbits later denoted by s_1 and t_1 . We compare the edges from E to F and to $-F$. For $E \leftrightarrow F$, the midpoint is

$$\frac{E + F}{2} = (-3, -4, -1) = -(3, 4, 1),$$

so its oriented primitive direction has key $(36, 3, 4, 1)$. For $E \leftrightarrow -F$, the midpoint is

$$\frac{E - F}{2} = (1, -1, -1),$$

whose key is $(5, 1, -1, -1)$. The second key is smaller. Therefore

$$\kappa([E], [F]) = (5, 1, -1, -1), \quad \eta([E], [F]) = -1,$$

and the corresponding map on representatives is

$$E \mapsto -F = (4, 3, 0).$$

For the other source point $-E$, multiplying both endpoints by -1 gives the other lift $-E \mapsto F$. The two source points are therefore paired with the two target points, and these pairings can be read in either direction.

If the actual source is σE , where $\sigma \in \{\pm 1\}$, the lift of an orbit edge is

$$(12) \quad \sigma E \mapsto \sigma \eta([E], [F]) F.$$

The same rule, read from the target, recovers the source.

Proposition 4.3 (Equal-key classes are matchings). *For a fixed key k , the edges of G_n with key k form a matching: no two share a source or target orbit.*

Proof. At a fixed signed endpoint, a primitive direction determines at most one opposite signed endpoint by Lemma 2.4. It remains to check that the two signs of an endpoint do not create two orbits. If a fixed direction sends F to $E = -F + 2qd$, then at the endpoint $-F$ the multiplier is $-q$ and the partner is

$$-(-F) + 2(-q)d = -E.$$

Thus changing the sign of one endpoint changes the sign of the other and does not change its orbit. Hence a fixed key occurs at most once at either endpoint of G_n . $\square$

Example 4.4 (Two disjoint edges with the same key). The key

$$(5, 1, -1, -1)$$

occurs on the edge (s_1, t_1) computed in Example 4.2. For a second edge, take

$$E' = (-2, 5, -2) = \text{pc}(s_3), \quad F' = (0, -3, 4) = \text{pc}(t_4).$$

Their midpoint is $(-1, 1, 1) = -(1, -1, -1)$, so this edge has the same key. The sources s_1, s_3 and targets t_1, t_4 are distinct. Hence the two edges with the same key are disjoint.

Every vertex ranks its incident edges by increasing κ . Proposition 4.3 implies that equal-key edges are vertex-disjoint, so every vertex has a strict preference order. A pair $(s, t) \notin M$ blocks a matching M if both s and t prefer each other to their partners in M , with being unmatched least preferred. A matching with no blocking pair is *stable*. Thus G_n is a stable-marriage instance with globally ranked pairs $[4, 5, 1]$: the two preference systems are induced by one global ranking of edges.

Proposition 4.5 (Unique stable matching). *The graph G_n has a unique stable matching. It is obtained by processing the edge keys in increasing order and selecting every edge whose two endpoints are still unmatched. Since $|\bar{\mathcal{S}}_n| = |\bar{\mathcal{T}}_n|$, this stable matching is perfect.*

Proof. First run the stated greedy procedure. If an edge is not selected, then at least one of its endpoints was already matched when that edge was processed. The selected edge at that endpoint has smaller key; equality is impossible by Proposition 4.3. Thus an unselected edge cannot block, and the greedy matching is stable. It is also perfect: otherwise balance would leave an unmatched source and an unmatched target, but the edge joining them would have been selected when it was processed.

It remains to prove uniqueness. Consider the least key among the edges whose endpoints have not already been removed. Every edge of that key is forced in any stable matching: neither endpoint has a smaller edge in the remaining graph, so omission would make the two endpoints a blocking pair. Proposition 4.3 says that these edges have disjoint endpoints, and hence they can be selected and removed simultaneously. A removed endpoint prefers its selected edge to every edge processed later, and the restriction of a stable matching to the remaining vertices is stable. Induction therefore shows that every stable matching agrees with the greedy matching. $\square$

Example 4.6 (A two-by-two globally ranked instance). Before returning to the arithmetic graph, consider two sources s_1, s_2 and two targets t_1, t_2 . Give the four edges the global ranks

	t_1	t_2
s_1	1	3
s_2	2	4

with smaller ranks preferred. The greedy rule first selects (s_1, t_1) . The edges of ranks 2 and 3 then meet an already matched vertex, so the next selected edge is (s_2, t_2) . Hence

$$M = \{(s_1, t_1), (s_2, t_2)\}.$$

The only other perfect matching is not stable, because (s_1, t_1) blocks it. This is the simplest instance of Proposition 4.5.

5. THE TWO-TO-ONE MAP

Let

$$\overline{M}_n : \overline{\mathcal{S}}_n \xrightarrow{\sim} \overline{\mathcal{T}}_n$$

be the unique stable matching of Proposition 4.5. Lift it to residual points using (12), and return to the original coordinates. Denote the resulting residual bijection by

$$\Psi_n^\circ : \mathcal{B}_{n,\circ}^{\text{odd}} \xrightarrow{\sim} \mathcal{A}_{n,\circ}.$$

Define the odd-branch bijection

$$\Psi_n(X) = \begin{cases} \Lambda_n(X), & X \in D_n, \\ \Psi_n^\circ(X), & X \in \mathcal{B}_{n,\circ}^{\text{odd}}. \end{cases}$$

Proof of Theorem 1.2. The map Λ_n is a bijection $D_n \rightarrow I_n$ by Theorem 3.2. The residual orbit matching is perfect by Proposition 4.5, and the sign lift pairs the two points in each source orbit with the two points in the matched target orbit. Hence Ψ_n is a bijection $\mathcal{B}_n^{\text{odd}} \rightarrow \mathcal{A}_n$.

Define

$$\Phi_n(x, y, z) = \begin{cases} (x, y, z/2), & z \text{ even}, \\ \Psi_n(x, y, z), & z \text{ odd}. \end{cases}$$

For $(u, v, w) \in \mathcal{A}_n$, the even branch contributes the preimage $(u, v, 2w)$, and the odd branch contributes the unique preimage $\Psi_n^{-1}(u, v, w)$. They are distinct because their third coordinates have opposite parity, and there are no further preimages on either branch. Thus every fiber has cardinality two. $\square$

Proof of Theorem 1.3. For an odd squarefree congruent number n , Tunnell's theorem gives (3), so Theorem 1.2 applies. $\square$

Remark 5.1 (Role of cardinality). The formulas defining the quarter-turns and edge keys make sense without (3). The equality is used only to make the two residual orbit sets equally large; Proposition 4.5 then makes their stable matching perfect. The later generator and deferred-acceptance procedure compute this already defined matching.

Example 5.2 (The direct $n = 41$ fiber). Let $Y = (2, -1, 1) \in \mathcal{A}_{41}$. The even branch gives

$$(2, -1, 2) \mapsto (2, -1, 1),$$

while Example 3.3 gives the odd preimage

$$(-4, -1, 1) \mapsto (2, -1, 1).$$

Their last coordinates have opposite parity, so

$$\Phi_{41}^{-1}(2, -1, 1) = \{(2, -1, 2), (-4, -1, 1)\}.$$

These are the two preimages listed in the first row of the introductory table.

6. ARITHMETIC GENERATION OF PREFERENCE LISTS

We now explain how to compute the residual matching without first constructing all m^2 edges. For a fixed source orbit, its preference list can be recovered by searching for the primitive midpoint directions of its incident edges. We first bound this search and then write it as a problem in a rank-two affine lattice.

Lemma 6.1 (Bound for the preferred direction). *If d is the direction realizing the smaller signed orientation of a residual orbit edge, then*

$$Q(d) \leq \frac{n-1}{2}.$$

Proof. The two midpoint decompositions from Lemma 4.1 give

$$E = qd + ce, \quad F = qd - ce.$$

Since d and e are orthogonal, we also have

$$n = q^2 Q(d) + c^2 Q(e).$$

The chosen orientation gives $\text{key}(d) < \text{key}(e)$, hence $Q(d) \leq Q(e)$. Since q and c are nonzero,

$$n \geq Q(d) + Q(e) \geq 2Q(d).$$

The claim follows because n is odd. $\square$

Example 6.2 (The midpoint bound is attained). In Example 2.5, the preferred direction satisfies

$$Q(d) = 20 = \frac{41-1}{2}.$$

Thus the bound in Lemma 6.1 can be an equality.

6.1. An orthogonal-complement lattice. We use the isometric embedding

$$\tau(X_1, X_2, X_3) = (X_1 + X_3, X_2, X_1 - X_3).$$

Then

$$\|\tau(X)\|^2 = Q(X), \quad \tau(X) \cdot \tau(Y) = \langle X, Y \rangle_Q.$$

The image of τ is the index-two lattice

(13)

$$\mathcal{L}_\tau = \{r \in \mathbb{Z}^3 : r_1 \equiv r_3 \pmod{2}\}, \quad \tau^{-1}(r) = \left(\frac{r_1 + r_3}{2}, r_2, \frac{r_1 - r_3}{2} \right).$$

Let us fix a residual source orbit. We write E for its canonical lifted representative and set $p = \tau(E)$. Since $\|p\|^2 = n$ and n is squarefree, p is primitive; otherwise the square of a common divisor of its coordinates would divide n .

We first describe which lattice points are needed. If $h = Q(d)$ and $t = qh$, then $r = \tau(d)$ is an integer point in

$$(14) \quad \underbrace{\{r : p \cdot r = t\}}_{\text{an affine plane}} \cap \underbrace{\{r : \|r\|^2 = h\}}_{\text{a sphere}} \cap \mathcal{L}_\tau.$$

The integer points parallel to this plane form the rank-two lattice $p^\perp \cap \mathbb{Z}^3$. We next choose coordinates on this lattice and use them to turn the plane-sphere intersection into a quadratic equation in two integers.

Lemma 6.3 (A basis of the orthogonal lattice). *There are deterministically computable vectors $e_1, e_2, z_0 \in \mathbb{Z}^3$ such that*

$$e_1 \times e_2 = p, \quad p \cdot z_0 = 1.$$

The pair e_1, e_2 is an oriented basis of $p^\perp \cap \mathbb{Z}^3$ and may be Lagrange–Gauss reduced while preserving $e_1 \times e_2 = p$.

Proof. Since p is primitive, there is a Bézout vector z_0 with $p \cdot z_0 = 1$. Applying Smith normal form to the row vector p gives an integral basis e_1, e_2 of its kernel. Every $x \in \mathbb{Z}^3$ has the unique decomposition

$$x = (x - (p \cdot x)z_0) + (p \cdot x)z_0.$$

Thus e_1, e_2, z_0 form a basis of $\mathbb{Z}^3$. The vector $e_1 \times e_2$ is an integral multiple of the primitive normal vector p . Taking its scalar product with z_0 shows that the multiple is ± 1 , and reversing the kernel basis if necessary gives $e_1 \times e_2 = p$. The usual integral Lagrange–Gauss steps

$$(e_1, e_2) \mapsto (e_1, e_2 - me_1), \quad (e_1, e_2) \mapsto (e_2, -e_1)$$

have determinant 1, preserve the cross product, and terminate with $|2e_1 \cdot e_2| \leq \|e_1\|^2 \leq \|e_2\|^2$. $\square$

We use Smith normal form in its usual algorithmic form: integer column operations, obtained from the Euclidean algorithm, change the primitive row p to $(1, 0, 0)$. Transforming the last two standard basis vectors back gives a basis of $p^\perp \cap \mathbb{Z}^3$. We take nonnegative Euclidean remainders and use lexicographic tie breaking. For Lagrange–Gauss reduction, we round ties in the nearest-integer step toward zero and choose the lexicographically smaller oriented basis when the two reduced vectors have the same norm. These choices make the basis and Bézout vector deterministic. They do not affect the final preference list, which is sorted by the intrinsic direction key. We refer to [3, Chap. 2] for these standard lattice algorithms.

We write

$$\begin{aligned} \alpha &= \|e_1\|^2, & \beta &= e_1 \cdot e_2, & \gamma &= \|e_2\|^2, \\ \delta &= z_0 \cdot e_1, & \varepsilon &= z_0 \cdot e_2, & \zeta &= \|z_0\|^2. \end{aligned}$$

The identity $e_1 \times e_2 = p$ gives

$$(15) \quad \alpha\gamma - \beta^2 = n.$$

This follows from

$$\alpha\gamma - \beta^2 = \|e_1 \times e_2\|^2 = \|p\|^2 = n.$$

6.2. Affine norm equations. For an edge incident to E , write its midpoint data as

$$\frac{E + F}{2} = qd, \quad h = Q(d), \quad r = \tau(d).$$

We then have

$$(16) \quad \|r\|^2 = h, \quad p \cdot r = qh.$$

We set $t = qh$. The frame from the previous subsection now gives coordinates on the affine plane $p \cdot r = t$.

Lemma 6.4 (Parametrization of a lattice coset). *Every integral solution of $p \cdot r = t$ is uniquely of the form*

$$r = tz_0 + \mu e_1 + \nu e_2, \quad \mu, \nu \in \mathbb{Z}.$$

For fixed h, q, ν , the condition $\|r\|^2 = h$ is equivalent to

$$(17) \quad \alpha\mu^2 + 2(\beta\nu + t\delta)\mu + \gamma\nu^2 + 2t\varepsilon\nu + t^2\zeta - h = 0.$$

Proof. The difference $r - tz_0$ is orthogonal to p , and e_1, e_2 form an integral basis of the orthogonal-complement lattice. Expanding the squared norm gives (17). $\square$

To determine the possible values of ν , we write

$$K = \beta\delta - \alpha\varepsilon, \quad M = h(n - q^2h).$$

The discriminant of (17), after dividing by 4, is

$$\Delta_\nu = (\beta\nu + t\delta)^2 - \alpha(\gamma\nu^2 + 2t\varepsilon\nu + t^2\zeta - h).$$

Lemma 6.5 (Bounds for the affine coordinates). *One has*

$$(18) \quad n\Delta_\nu = \alpha M - (n\nu - tK)^2.$$

Put $R = \lfloor \sqrt{\alpha M} \rfloor$. By (18), $\Delta_\nu \geq 0$ exactly when ν lies in the interval

$$(19) \quad \left\lceil \frac{tK - R}{n} \right\rceil \leq \nu \leq \left\lfloor \frac{tK + R}{n} \right\rfloor.$$

For each such ν , let $s = \lfloor \sqrt{\Delta_\nu} \rfloor$. If $s^2 = \Delta_\nu$, the possible values of μ form the set

$$(20) \quad \left\{ \frac{-(\beta\nu + t\delta) + u}{\alpha} : u \in \{s, -s\}, \quad \alpha \mid -(\beta\nu + t\delta) + u \right\}.$$

If $s^2 \neq \Delta_\nu$, there are no roots. In particular, when $\Delta_\nu = 0$, the set (20) contains at most one integer.

Proof. The Gram matrix of the basis e_1, e_2, z_0 is

$$G = \begin{pmatrix} \alpha & \beta & \delta \\ \beta & \gamma & \varepsilon \\ \delta & \varepsilon & \zeta \end{pmatrix}.$$

Since $\det[e_1 \ e_2 \ z_0] = 1$, one has $\det G = 1$. Expanding gives

$$n\zeta - (\alpha\varepsilon^2 - 2\beta\delta\varepsilon + \gamma\delta^2) = 1,$$

and a direct rearrangement using $n = \alpha\gamma - \beta^2$ gives

$$K^2 + n(\delta^2 - \alpha\zeta) = -\alpha.$$

The discriminant can now be written as

$$\Delta_\nu = -n\nu^2 + 2tK\nu + t^2(\delta^2 - \alpha\zeta) + \alpha h.$$

After multiplying by n and using the preceding identity together with $t = qh$, we obtain

$$n\Delta_\nu = \alpha(nh - t^2) - (n\nu - tK)^2 = \alpha h(n - q^2h) - (n\nu - tK)^2,$$

which is (18). Since $n\nu - tK$ is integral, its square is at most αM precisely when $|n\nu - tK| \leq \lfloor \sqrt{\alpha M} \rfloor$; this is equivalent to (19). The quadratic formula gives (20), interpreted as a set. $\square$

Example 6.6 (Recovering one $n = 41$ edge arithmetically). Return to $E = (-2, -5, 2)$ and $F = (4, -3, 0)$ from Example 2.5. The isometry gives

$$p = \tau(E) = (0, -5, -4).$$

One convenient oriented kernel basis and Bézout vector are

$$e_1 = (1, 0, 0), \quad e_2 = (0, -4, 5), \quad z_0 = (0, -1, 1).$$

These vectors follow directly from the two defining equations. The kernel equation $p \cdot r = 0$ is $-5r_2 - 4r_3 = 0$, so $r_2 = -4k$ and $r_3 = 5k$, while r_1 is free. This gives e_1, e_2 , and the Bézout identity $5 - 4 = 1$ gives z_0 . Hence $e_1 \times e_2 = p$ and $p \cdot z_0 = 1$. For this frame,

$$\begin{aligned} \alpha &= 1, & \beta &= 0, & \gamma &= 41, \\ \delta &= 0, & \varepsilon &= 9, & \zeta &= 2, & K &= -9. \end{aligned}$$

The midpoint direction is $d = (1, -4, 1)$, so

$$h = Q(d) = 20, \quad q = 1, \quad t = 20, \quad M = 20(41 - 20) = 420.$$

Since $R = \lfloor \sqrt{420} \rfloor = 20$ and $tK = -180$, the interval (19) becomes

$$\left\lceil \frac{-200}{41} \right\rceil \leq \nu \leq \left\lfloor \frac{-160}{41} \right\rfloor,$$

which leaves only $\nu = -4$. The discriminant identity then gives

$$41\Delta_{-4} = 420 - (41(-4) - 20(-9))^2 = 164, \quad \Delta_{-4} = 4.$$

Thus (20) gives $\mu = \pm 2$. The choice $\mu = 2$ produces

$$r = 20z_0 + 2e_1 - 4e_2 = (2, -4, 0), \quad \tau^{-1}(r) = (1, -4, 1) = d.$$

The other root has negative first coordinate and is removed by the orientation convention. The outer loop also uses negative q , so the canonically oriented

opposite direction is recovered there whenever it gives the preferred signed edge. Finally,

$$-E + 2d = (4, -3, 0) = F.$$

Since the canonical representative of $[F]$ is $F_0 = (-4, 3, 0)$, the generator records

$$((20, 1, -4, 1), [F_0], -1).$$

This recovers one entry of the preference list directly from the arithmetic of the source E .

6.3. Generating a preference list. The preceding calculation gives a finite procedure for producing the complete preference list of a fixed source orbit.

Algorithm 6.7 (Generation of a preference list). Fix a residual source orbit $[E]$ and let $E = \text{pc}([E])$. We perform the following steps:

- (1) let $h = 1, 2, \dots, (n-1)/2$;
- (2) let q range over the nonzero odd integers satisfying $q^2 h < n$, and put $t = qh$ and $M = h(n - q^2 h)$;
- (3) let ν range over the interval (19);
- (4) compute Δ_ν and the set of integral roots (20); for each root form $r = tz_0 + \mu e_1 + \nu e_2$;
- (5) retain r only if $r_1 \equiv r_3 \pmod{2}$ and $d = \tau^{-1}(r)$ is primitive, canonically oriented, and has odd third coordinate;
- (6) form $F = -E + 2qd$ and retain it only if $F \in \mathcal{T}_n^\circ$; write $F = \eta F_0$, where $F_0 = \text{pc}([F])$ and $\eta \in \{\pm 1\}$;
- (7) compare d with the primitive oriented direction of $(E - F)/2$. Retain the record

$$(\text{key}(d), [F_0], \eta)$$

precisely when d has the smaller key.

For each h , we place the surviving records in a finite set and sort them by (d_1, d_2, d_3) . Since h increases in the outer loop, these blocks together form the strict preference list of $[E]$. The square roots and divisibility tests use only integer arithmetic. When the discriminant is zero, the two signs in the quadratic formula contribute only one root. The generator keeps one sorted h -block at a time, returns its records, and then moves to the next nonempty block.

Theorem 6.8 (Correctness of the preference-list algorithm). *For every residual source orbit $[E]$, Algorithm 6.7 terminates and outputs every incident edge exactly once, in strictly increasing order of its key.*

Proof. We first check that every output is an incident edge. A retained record satisfies $Q(d) = h$ and $\langle E, d \rangle_Q = qh$, and hence

$$Q(-E + 2qd) = Q(E) - 4q \langle E, d \rangle_Q + 4q^2 Q(d) = n.$$

The test $F \in \mathcal{T}_n^\circ$ gives the required congruence and exclusion conditions, so the record defines a valid edge between the chosen representatives. The last

comparison keeps the preferred sign of the target orbit, and η is the sign used in (12).

Conversely, take an incident edge and choose its preferred signed lift from E . Lemma 6.1 gives the stated range for $h = Q(d)$. Its vectors p and $r = \tau(d)$ satisfy (16). By Lemma 6.4, r has unique integral coordinates μ, ν , and Lemma 6.5 shows that the procedure recovers them. The τ -lattice, primitivity, orientation, parity, and residual tests all hold for this lift, and the final comparison retains it.

It remains to check that no edge appears twice. The values h and d are intrinsic to a retained edge, while $q = \langle E, d \rangle_Q / Q(d)$ is then forced. The affine coordinates μ, ν are unique, and (20) is a set even when $\Delta_\nu = 0$. Hence no record is emitted twice. $\square$

Example 6.9 (A complete preference list at $n = 41$). For $E = (-2, -5, 2)$, direct enumeration of the residual target set gives four target orbits. Write their canonical representatives as

$$\begin{aligned} F_1 &= (-4, -3, 0), & F_2 &= (0, -3, 4), \\ F_3 &= (-4, 3, 0), & F_4 &= (0, -3, -4). \end{aligned}$$

Algorithm 6.7 selects the following midpoint for each orbit. The last column records the Q -value of the discarded midpoint direction obtained from the opposite sign.

orbit	η	$(E + \eta F_i)/2$	$\kappa([E], [F_i])$	other Q
$[F_1]$	-1	$(1, -1, 1)$	$(5, 1, -1, 1)$	36
$[F_2]$	-1	$(-1, -1, -1)$	$(5, 1, 1, 1)$	36
$[F_3]$	-1	$(1, -4, 1)$	$(20, 1, -4, 1)$	21
$[F_4]$	1	$(-1, -4, -1)$	$(20, 1, 4, 1)$	21

The corresponding affine data (h, q, μ, ν) , in the same row order, are

$$(5, 1, 2, -1), \quad (5, -1, 2, 1), \quad (20, 1, 2, -4), \quad (20, -1, 2, 4).$$

For example, the third row is the full calculation in Example 6.6; its opposite sign gives the direction $(3, 1, -1)$ of Q -value 21. In the second and fourth rows the displayed midpoint is the negative of its canonically oriented primitive direction, which is why the key has positive first coordinate. Sorting the surviving records gives the preference list

$$[F_1], [F_2], [F_3], [F_4]$$

with keys in the order displayed. Each of the four target orbits occurs once.

Remark 6.10 (A bound from lattice reduction). For a Lagrange–Gauss reduced basis,

$$|2\beta| \leq \alpha \leq \gamma.$$

Equation (15) gives $\alpha \leq 2\sqrt{n/3}$. The interval (19) therefore has a length bounded in terms of n and M . We do not need this bound for the correctness of the algorithm or for the proposal counts below.

7. DEFERRED ACCEPTANCE WITH GENERATED PREFERENCES

Proposition 4.5 describes the matching by a global greedy procedure. To avoid sorting all edge keys at once, we use the source-proposing Gale–Shapley algorithm [4] together with the preference lists above. We compute entries as they are requested and retain the position reached in each list. The order of the preferences remains fixed throughout the algorithm.

Enumerate the finite residual orbit sets $\overline{\mathcal{S}}_n$ and $\overline{\mathcal{T}}_n$, and attach to each source $s \in \overline{\mathcal{S}}_n$ the ordered generator of Theorem 6.8. A target compares two proposals by their global edge rank.

Algorithm 7.1 (Source-proposing deferred acceptance). Initially every source is free and every target holds no proposal. Place the sources in a first-in, first-out queue in lexicographic order of their canonical representatives. Repeatedly remove a free source s from the front of the queue and request the next record (k, t, η) from its ordered generator.

- (1) If t holds no proposal, it holds (s, k, η) .
- (2) If t holds (s', k', η') and $k < k'$, then t replaces that record by (s, k, η) and s' is appended to the queue.
- (3) Otherwise t rejects s , which is appended to the queue.

The algorithm ends when the queue is empty.

Example 7.2 (Deferred acceptance on two sources). Run Algorithm 7.1 on the ranks in Example 4.6, with initial queue (s_1, s_2) . The proposals are

$$s_1 \longrightarrow t_1, \quad s_2 \longrightarrow t_1 \text{ (rejected)}, \quad s_2 \longrightarrow t_2.$$

The first proposal is held by t_1 . The second has rank 2, so t_1 keeps its rank-1 proposal from s_1 and rejects s_2 . Source s_2 then proposes to t_2 . The queue is empty and the output is $\{(s_1, t_1), (s_2, t_2)\}$, the unique stable matching found by the greedy argument.

Theorem 7.3 (Correctness of deferred acceptance). *Algorithm 7.1 terminates and returns the unique perfect stable matching of Proposition 4.5. The stored sign η determines the paired representatives through (12). The inverse at a target is recovered by rerunning the same deterministic matching algorithm; the sequence of earlier proposals need not be retained.*

Proof. A source proposes in strict preference order and never proposes to the same target twice. We first check that a free source never exhausts its generator. If a free source s had proposed to all m targets, then every target would hold a proposal: once a target receives its first proposal, it remains occupied thereafter. The held proposals would come from m distinct sources, since a source can be held by at most one target and proposes again only after becoming free. None could come from the free source s , leaving only $m - 1$ possible sources, a contradiction. Thus every request made by the algorithm has a next record. Since there are only m^2 possible proposals, the algorithm terminates.

When the queue is empty, every source is held by a target. There are m sources and m targets, and no target holds two sources, so the resulting matching is perfect.

Suppose (s, t) blocked the final matching. Since s prefers t to its final partner, s proposed to t earlier. The target either rejected s immediately or later displaced it for a proposal of smaller key. A target's held key only decreases, so its final edge is smaller than (s, t) , a contradiction. The output is stable and hence equals the unique stable matching of Proposition 4.5.

For the inverse, let F_0 be the canonical representative of a matched target orbit, and suppose its matched record is (E_0, F_0, η) with E_0 canonical. The lifted matching sends E_0 to ηF_0 . Hence an actual target ρF_0 , with $\rho \in \{\pm 1\}$, has preimage $\rho \eta E_0$. Lemma 4.1 makes η intrinsic to the two orbits, and rerunning the deterministic computation reconstructs the same matching and the same sign. This recovers the inverse at the given target without retaining the first run's proposal transcript. $\square$

Example 7.4 (Recovering a residual preimage). Take the actual lifted target $F = (4, -3, 0)$. Its canonical orbit representative is $F_0 = (-4, 3, 0)$, so $F = \rho F_0$ with $\rho = -1$. Rerunning the deterministic matching returns the record from Example 6.9 with

$$E_0 = (-2, -5, 2), \quad \eta = -1.$$

The inverse sign formula gives

$$\rho \eta E_0 = (-1)(-1)E_0 = E_0.$$

Finally, $\iota_B^{-1}(E_0) = (-2, -5, 1)$. Thus the inverse at the original target is

$$(4, -3, 0) \mapsto (-2, -5, 1).$$

This inverse calculation requires rerunning the matching on all residual orbits before choosing the sign. It is not a formula depending only on the coordinates of the given target.

8. THE MAP AT $n = 41$

The quarter-turn calculation in Examples 3.3 and 5.2 gives

$$(-4, -1, 1) \mapsto (2, -1, 1) \quad \text{with even companion } (2, -1, 2).$$

We now work out the residual example

$$(-2, -5, 1) \mapsto (4, -3, 0) \quad \text{with even companion } (4, -3, 0).$$

In each case, the even preimage is obtained by doubling the third coordinate of the target.

Example 8.1 (The residual matching for $n = 41$). The residual graph for $n = 41$ has four source orbits $s_1, \dots, s_4$ and four target orbits $t_1, \dots, t_4$. We

index them by the following canonical lifted representatives:

$$\begin{aligned} \text{pc}(s_1) &= (-2, -5, -2), & \text{pc}(s_2) &= (-2, -5, 2), \\ \text{pc}(s_3) &= (-2, 5, -2), & \text{pc}(s_4) &= (-2, 5, 2), \\ \text{pc}(t_1) &= (-4, -3, 0), & \text{pc}(t_2) &= (-4, 3, 0), \\ \text{pc}(t_3) &= (0, -3, -4), & \text{pc}(t_4) &= (0, -3, 4). \end{aligned}$$

In the next display, each row is in preference order and the bracketed four-tuple is the corresponding edge key:

$$\begin{aligned} s_1 : & \quad t_1[5, 1, -1, -1], \quad t_3[5, 1, 1, -1], \quad t_2[20, 1, -4, -1], \quad t_4[20, 1, 4, -1], \\ s_2 : & \quad t_1[5, 1, -1, 1], \quad t_4[5, 1, 1, 1], \quad t_2[20, 1, -4, 1], \quad t_3[20, 1, 4, 1], \\ s_3 : & \quad t_4[5, 1, -1, -1], \quad t_2[5, 1, 1, -1], \quad t_3[20, 1, -4, -1], \quad t_1[20, 1, 4, -1], \\ s_4 : & \quad t_3[5, 1, -1, 1], \quad t_2[5, 1, 1, 1], \quad t_4[20, 1, -4, 1], \quad t_1[20, 1, 4, 1]. \end{aligned}$$

The preference list in Example 6.9 is the s_2 row here: its F_1, F_2, F_3, F_4 are respectively t_1, t_4, t_2, t_3 . With the queue convention of Algorithm 7.1, the proposal sequence is

$$\begin{aligned} s_1 &\longrightarrow t_1, & s_2 &\longrightarrow t_1 \quad (\text{rejected}), & s_3 &\longrightarrow t_4, \\ s_4 &\longrightarrow t_3, & s_2 &\longrightarrow t_4 \quad (\text{rejected}), & s_2 &\longrightarrow t_2. \end{aligned}$$

Thus the algorithm makes six proposals. Its lifted residual pairs are

$$\begin{aligned} (-2, -5, -2) &\longleftrightarrow (4, 3, 0), \\ (-2, -5, 2) &\longleftrightarrow (4, -3, 0), \\ (-2, 5, -2) &\longleftrightarrow (0, -3, 4), \\ (-2, 5, 2) &\longleftrightarrow (0, -3, -4). \end{aligned}$$

For instance,

$$(-2, -5, 1) \in \mathcal{B}_{41}^{\text{odd}} \longmapsto (4, -3, 0) \in \mathcal{A}_{41}.$$

Thus

$$\Phi_{41}^{-1}(4, -3, 0) = \{(-2, -5, 1), (4, -3, 0)\}.$$

9. PARKING FUNCTIONS AND THE COST OF DEFERRED ACCEPTANCE

A sequence $(a_1, \dots, a_m)$ of positive integers is a *parking function* if its nondecreasing rearrangement $b_1 \leq \dots \leq b_m$ satisfies $b_i \leq i$ for $1 \leq i \leq m$; see [7, 8]. For $s \in \mathcal{S}_n$, let $p(s)$ be the number of proposals made by s . Equivalently, $p(s)$ is the rank of its final target in its preference list. In the proof below, $M = \overline{M}_n$ denotes the final stable matching.

Theorem 9.1 (The proposal counts form a parking function). *Order the sources $s_1, \dots, s_m$ so that the keys of their final matched edges are nondecreasing. Then*

$$p(s_i) \leq i \quad (1 \leq i \leq m).$$

Let $b_1 \leq \dots \leq b_m$ be the nondecreasing rearrangement of the proposal counts. Then $b_i \leq i$, so the proposal-count sequence is a parking function. Moreover,

$$(21) \quad \sum_{s \in \bar{\mathcal{S}}_n} p(s) \leq \frac{m(m+1)}{2}.$$

Proof. Fix a source s , and let t be a target that s prefers to its final target. Stability forces

$$\kappa(M^{-1}(t), t) < \kappa(s, t) < \kappa(s, M(s));$$

otherwise (s, t) would block. Distinct preferred targets give distinct matched edges of key strictly smaller than the matched edge of s . Hence

$$p(s) - 1 \leq \#\{e \in M : \kappa(e) < \kappa(s, M(s))\}.$$

For $s = s_i$, the right-hand side is at most $i - 1$, proving $p(s_i) \leq i$. For each i , the first i sources in this ordering provide i proposal counts, all at most i . Hence the i th smallest proposal count is at most i , which is the parking-function condition. Summing the inequalities $p(s_i) \leq i$ gives (21). $\square$

Example 9.2 (The parking function for $n = 41$). The residual run makes the six proposals

$$s_1 \rightarrow t_1, \quad s_2 \rightarrow t_1, \quad s_3 \rightarrow t_4, \quad s_4 \rightarrow t_3, \quad s_2 \rightarrow t_4, \quad s_2 \rightarrow t_2.$$

Thus

$$p(s_1) = 1, \quad p(s_2) = 3, \quad p(s_3) = 1, \quad p(s_4) = 1.$$

Its nondecreasing rearrangement is $(1, 1, 1, 3)$, and

$$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 3 \leq 4.$$

Thus it is a parking function. The total number of proposals is 6, while Theorem 9.1 gives the bound $4 \cdot 5/2 = 10$. The algorithm requests six preference-list entries and processes six proposals. Exactly two of these proposals are compared with a proposal already held by the target. These are the operations counted in Theorem 9.5.

Proposition 9.3 (Sharpness). *The bound (21) is sharp for the class of globally ranked-pairs instances, even when every key class consists of a single edge.*

Proof. Let the sides be $\{s_1, \dots, s_m\}$ and $\{t_1, \dots, t_m\}$, and give (s_i, t_j) the distinct label $(m+1)i + j$. Every source orders the targets as $t_1, t_2, \dots, t_m$, and every target orders the sources as $s_1, s_2, \dots, s_m$. In increasing-label greedy order, (s_1, t_1) is selected first; after its endpoints are removed, (s_2, t_2) is selected, and induction gives the diagonal stable matching. The final target t_i has rank i in the list of s_i , so s_i makes exactly i proposals. The total is $m(m+1)/2$. $\square$

Example 9.4 (Sharpness with three sources). For $m = 3$, all sources rank the targets as t_1, t_2, t_3 , and all targets rank the sources as s_1, s_2, s_3 . Starting with the queue (s_1, s_2, s_3) , source s_1 is held by t_1 . Sources s_2 and s_3 are then rejected by t_1 ; next s_2 is held by t_2 , and s_3 is rejected by t_2 before reaching t_3 . The proposal counts are therefore

$$p(s_1) = 1, \quad p(s_2) = 2, \quad p(s_3) = 3,$$

whose total $6 = 3 \cdot 4/2$ attains the bound.

Theorem 9.5 (Cost of deferred acceptance). *Let*

$$N = \sum_{s \in \bar{S}_n} p(s)$$

be the number of proposals. With preference lists supplied by the generators of Theorem 6.8, the deferred-acceptance algorithm makes N requests for the next list entry, performs $O(N)$ key comparisons and queue or assignment operations, and stores $O(m)$ auxiliary records apart from those stored by the generators. In particular,

$$N \leq \frac{m(m+1)}{2}.$$

The stated bound concerns the deferred-acceptance phase only. It does not include the arithmetic work or storage internal to the generators, the enumeration of the residual orbits, or integer bit complexity.

Proof. Processing a proposal requires one key comparison if the target already holds a proposal, and none otherwise. It also requires a constant number of queue and matching-array updates. Apart from the generators, the algorithm stores the status of each source, one held record per target, and the queue of free sources. These require $O(m)$ records. The bound on N is Theorem 9.1. $\square$

The theorem does not bound the total cost of evaluating Φ_n : producing the next preference-list entry may require further arithmetic. Computing an inverse also requires another run of the matching algorithm, whose cost is not included above.

Remark 9.6 (On possible subquadratic bounds). Proposition 9.3 shows that an $O(m \log m)$ worst-case bound on the number of proposals cannot follow from the globally-ranked-pairs property alone. Such a bound for the Tunnell-form instances considered here would require additional arithmetic information about the distribution of their midpoint directions.

10. CONCLUDING REMARKS

We have constructed the odd-branch bijection using three quarter-turns and a stable matching on the remaining antipodal orbits. Primitive midpoint directions determine the preferences, and the common ranking of edges gives both uniqueness and the parking-function bound. The examples at $n = 41$

show the two cases: an odd preimage found by a quarter-turn and one found by the residual matching. In both cases, doubling the third coordinate of the target gives the even preimage.

The preference lists can be generated without storing the complete $m \times m$ edge table. It remains to determine whether the arithmetic distribution of midpoint directions gives a better complexity bound for these representation sets. A separate question is whether the residual partner can be found directly from one representation, without computing the global matching.

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DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

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